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APPENDIX QC: QUANTUM COMPUTING — TRIADIC NISQ ARCHITECTURE

Solving the Noisy Intermediate-Scale Quantum Problem via U-Theory & Lady Galaxy Protocol

Version 25.2 — THE QUANTUM COMPUTING EDITION

Copyright c 2026 Petar Nikolov. All rights reserved. Licensed under CC BY 4.0. Standalone appendix for U-Theory / U-Model v25
Status: L2 STRUCTURAL ISOMORPHISM + L3 ENGINEERING APPLICATION Last Updated: February 21, 2026 Epistemic Level: L2
70–90 for triadic mapping; $L3 < 50$ for speculative predictions DOI: [10.17605/OSF.IO/74XGR](https://doi.org/10.17605/OSF.IO/74XGR) | Zenodo: [10.5281/zenodo.18475832](https://zenodo.org/record/18475832)
Prerequisites: Appendix QM Quantum Mechanics Application, Appendix RR Resistance Model, Appendix LG
Lady Galaxy Protocol, Appendix RP Triadic Research Law

Changelog v25.0 *Swan11*:

Change	Description
+ QC.12	Center-Periphery distribution in quantum chips <i>direct from v24.4</i>
+ QC.13	Mirror Entanglement as Shared Form <i>Mirror Theory integration</i>
+ QC.14	Triadic Quantum Internet Protocol <i>QIP – T</i>
+ QC.15	U-Score as official NISQ benchmarking standard <i>proposal</i>
+ QC.16	Triadic Path to Fault-Tolerant QC <i>vs brute – force QEC roadmap</i>
+ QC.17	L3 Speculative: X-Category residue in superconducting hardware
+ QC.18	12 new predictions + 2026–2028 experimental roadmap
+ QC.20	Triadic Error Mitigation <i>ZNE, PEC, CDR as F – P – A strategies</i>
+ QC.21	Barren Plateaus as Triadic Death <i>VQA gradient catastrophe + structured ansatz</i>
QC.9 updated	Predictions extended to QC-P1...QC-P17
QC.10 updated	Swan10 + Swan11 honest assessment
QC.11 updated	7 new references 20–26

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HIERARCHY REMINDER

Layer	Scope	This Appendix Covers
U-Theory	Universe $L1 + L2$	Triadic mapping of quantum hardware <i>Form–Position–Action</i> in qubits
U-Model	Earth $L3$	Engineering protocol for NISQ optimization via Lady Galaxy Protocol
U-Score	Metrics	U_{triad} , δ -imbalance, Coherence Efficiency Ratio for quantum circuits

"The quantum computer is the most fragile possible material realization of the Triad at macro scale. It breaks not because physics is cruel — but because the industry commits the Sisyphus Error."

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QC.0: Executive Summary — The NISQ Crisis as Triadic Imbalance

The Problem

The quantum computing industry 2020–2026 suffers from a classical error defined in U-Theory as **The Sisypheus Error** *Appendix RP.4*:

***Symptom:** Trying to compensate for weak **Form** coherence and poor **Position** connectivity through massive amounts of **Action** number of physical qubits and quantum gates.*

***U-Model Diagnosis:** $\delta \rightarrow 1$ maximum triadic imbalance. The Stability Index collapses: $SI = U / (1 + \delta)^2 \rightarrow 0$.*

The industry's official doctrine — **Brute-Force Quantum Error Correction QEC** — demands ~10,000 noisy physical qubits to assemble 1 stable logical qubit. This is an attempt to compensate Form with monstrous expenditure of Space and Energy. This path will take decades.

The Solution *This Appendix*

The **Lady Galaxy Protocol** *Map* $\rightarrow$ *Standardize* $\rightarrow$ *Pulse* decomposes this complex physical problem along the three axes and provides an **architectural solution** through the Three Prices of Existence:

$$T_{\text{coherence}} \propto \frac{\rho_D^{\text{device}}}{Z_A^{\text{environment}}} \cdot S_P$$

This is **Prediction DP-S6** from U-Theory: coherence time is proportional to Form strength ρ_D , inversely proportional to environmental Action impedance Z_A , and scaled by Position topology S_P .

Instead of more hardware *raw bricks / raw energy*, the solution is **orthogonal reduction of all three resistances** through reformatting of software logic.

QC.1: Phase 1 — Quantum Triadic Mapping *LGP* – 1

Protocol Step: LGP-1TriadicIntake — Map the system onto Form–Position–Action.

A quantum computer is not merely a computational machine; it is the **most fragile possible material realization of the Triad** at macro-level:

QC.1.1: The Canonical Mapping

Triad <i>U – Theory</i>	Mirror / Resistance	In Quantum Computing	Price You Pay	Role in U-Model
FORM <i>F</i>	TIME \$ ρ_D \$	Superposition — the wave function ψ . The identity of the quantum state before measurement.	Durability — coherence time T_1/T_2 . You pay with lifespan to keep the state from decaying <i>Decoherence</i> .	Form protects information from Chaos. Without it → classical noise.
POSITION <i>P</i>	SPACE \$ R_P \$	Connectivity / Entanglement Topology — the geometry by which qubits can interact.	Distance/Transfer — SWAP gates. This is structural friction.	Position gives Context. A qubit alone is worthless; its value lies in the entanglement network.
ACTION <i>A</i>	ENERGY \$ Z_A \$	Quantum Gates operators and Measurement — CNOT, Hadamard, and collapse of the wave function.	Entropy — A_{loss} <i>heat, errors, environmentalnoise</i> at every state transition.	Action does the work, but " <i>Every error is a small death.</i> "

QC.1.2: Energy Balance

Every quantum operation obeys the U-Theory energy equation:

$$A_{\text{in}} = A_{\text{eff}} + A_{\text{loss}}$$

Where: - A_{in} = total energy/action invested *gatepulses, cooling, controlsignals* - A_{eff} = useful computational work *entanglementcreated, logicperformed* - A_{loss} = quantum noise, heating, decoherence — the **entropy tax**

The **Action Efficiency** is:

$$\eta_A = \frac{M_{\text{meaning}}}{M + \mathcal{W}_{\text{waste}}}$$

where M_{meaning} is the meaningful information extracted, M is the total measurement, and $\mathcal{W}_{\text{waste}}$ is wasted entropy.

QC.1.3: Why Quantum Computing is the Ultimate Triadic Test

Scale	System	Form Stability	Position Stability	Action Stability
Macro <i>Classical</i>	Steel bridge	Years	Fixed	Predictable
Meso <i>Biological</i>	Living cell	Hours–days	Bounded	Controlled
Micro <i>Quantum</i>	Superconducting qubit	~50–200 μs	2–4 neighbors	99.0–99.9% fidelity

The quantum computer operates at the **absolute boundary** where triadic stability is barely possible. This makes it the most demanding — and the most revealing — test of U-Theory's predictions.

QC.2: Phase 2 — Resistance Report *LGP* — 2

Protocol Step: LGP-2 ResistanceReport — Diagnose where the "glass cup" breaks.

QC.2.1: Resistance Formula

$$\mathcal{R}(\Pi_{\text{NISQ}}) = \{R_P^{\text{high}}, \rho_D^{\text{low}}, Z_A^{\text{high}}\}$$

All three resistances are critical. This is why NISQ is so hard — there is no single bottleneck, but a **triadic crisis**.

QC.2.2: Resistance #1 — High Position Inertia $\$R_P\$$

Diagnosis: We pay too high a price for Space.

Reality: Physical qubits in today's chips *IBMEagle/Heron, GoogleSycamore* typically communicate with only **2 to 4 nearest neighbors** *2Dlattice* topology.

LGP Diagnosis: If the algorithm requires qubit 1 to interact with qubit 50, you must execute a chain of **SWAP operations**. Each SWAP: - Consumes time *eats into Form's lifetime* - Injects noise $\$A_{\text{loss}}\$$ *increases* - The geometry destroys the result

Quantification:

$$\text{SWAP overhead} = O(d_{\text{graph}}(q_i, q_j))$$

where d_{graph} is the graph distance on the chip topology. For a 127-qubit IBM Eagle *heavy-hex*, the worst-case is $d \approx 20$ SWAPs, each with $\sim 0.5\text{--}1\%$ error.

Cumulative damage: $P_{\text{success}} = (1 - \epsilon_{\text{SWAP}})^{n_{\text{SWAP}}} \approx e^{-n \cdot \epsilon}$

For 20 SWAPs at 1% error: $P_{\text{success}} \approx 0.82$. **18% of the computation is lost to geometry alone.**

QC.2.3: Resistance #2 — Weak Form $\$\rho_D\$$

Diagnosis: The time for existence is critically short.

Reality: Thermal fluctuations, cosmic rays, and electromagnetic noise constantly "probe" the qubits. They collapse back to classical 0 or 1 within microseconds.

Platform	T_1 relaxation	T_2 dephasing	Gate time
IBM Heron <i>superconducting</i>	$\sim 300\ \mu\text{s}$	$\sim 200\ \mu\text{s}$	$\sim 30\text{--}60\ \text{ns}$
Google Sycamore <i>superconducting</i>	$\sim 20\ \mu\text{s}$	$\sim 10\ \mu\text{s}$	$\sim 12\text{--}25\ \text{ns}$
Quantinuum H2 <i>trapped ion</i>	$\sim 10\ \text{s}$	$\sim 1\ \text{s}$	$\sim 200\ \mu\text{s}$
IonQ Forte <i>trapped ion</i>	$\sim 10\ \text{s}$	$\sim 1\ \text{s}$	$\sim 600\ \mu\text{s}$

LGP Diagnosis: Form is too weak to withstand the Entropy of Action $\$Z_A\$$ from the environment.

Critical ratio *operations before death*:

$$N_{\text{ops}} = \frac{T_2}{t_{\text{gate}}} \approx \begin{cases} 3,000\text{--}6,000 & \text{(superconducting)} \\ 5,000\text{--}15,000 & \text{(trapped ion)} \end{cases}$$

This is the **maximum circuit depth** before decoherence kills the computation.

QC.2.4: Resistance #3 — High Action Impedance $\$Z_A\$$

Diagnosis: The "Action tax" leads to cumulative errors.

Reality: There are no ideal quantum gates. Current best fidelities:

Gate Type	Typical Fidelity	Error per gate
Single-qubit <i>superconducting</i>	99.95%	0.05%
Two-qubit CNOT <i>superconducting</i>	99.5%	0.5%
Two-qubit <i>trappedion</i>	99.8%	0.2%
Measurement	99.5%	0.5%

Cumulative collapse: For a circuit with n gates:

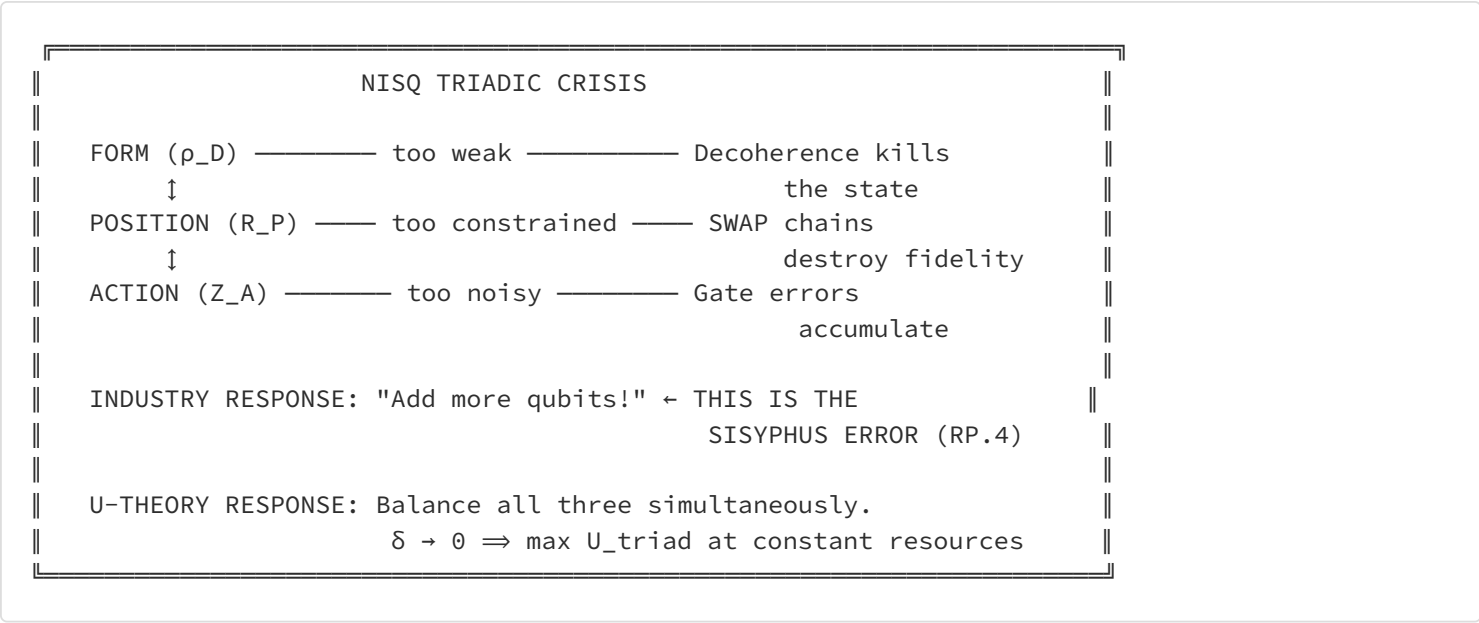
$$P_{\text{success}} = \prod_{i=1}^n (1 - \epsilon_i) \approx e^{-\sum \epsilon_i}$$

For 1,000 two-qubit gates at 0.5% error:

$$P_{\text{success}} \approx e^{-5} \approx 0.007$$

Only 0.7% chance of getting the right answer. This is the A_{loss} catastrophe from U-Theory.

QC.2.5: The Triadic Crisis Diagram



QC.3: Phase 3 — Triadic Synthesis of the Solution *LGP* — 6/8

Protocol Step: LGP-6 Synthesize → LGP-8 Pulse — Apply the triadic rebalancing.

The official doctrine today is Brute-Force QEC. U-Theory offers a fundamentally different path: **simultaneous orthogonal optimization of all three axes.**

QC.3.1: The Three-Axis Solution Matrix

Axis	Solution	Mechanism	U-Theory Principle	Expected Gain
F-Axis <i>Form</i>	Hyper-Form: DFS + Dynamical Decoupling	$E \perp \Gamma$ — encode in noise-orthogonal subspaces	Form protects information via orthogonality	10×+ coherence extension
P-Axis <i>Position</i>	Orthogonal Compiler: topology-aware transpilation	Metric Engineering via Least Action	Minimize R_P by mapping algorithm geometry to device geometry	30–60% SWAP reduction
A-Axis <i>Action</i>	Variational Hybrid: ultra-short quantum circuits	Minimize Z_A exposure via classical co-processing	Extract value before entropy accumulates	5–20× depth reduction

QC.3.2: The AM-GM Theorem Applied to NISQ

From the **Triadic Resonance Theorem** *Theorem*2, $U - Theory$ v18.0:

$$\max U_{\text{triad}} \text{ at constant total resource } R \iff \delta \rightarrow 0$$

Where δ is the triadic imbalance:

$$\delta = \frac{\max(F, P, A) - \min(F, P, A)}{\max(F, P, A)}$$

Translation for NISQ: Given a fixed number of noisy qubits: - Spending all resources on more qubits *Action* while ignoring coherence *Form* and connectivity *Position* $\rightarrow \delta \rightarrow 1 \rightarrow SI \rightarrow 0$ - **Balanced allocation** across all three axes $\rightarrow \delta \rightarrow 0 \rightarrow SI \rightarrow U \text{ maximum}$

The **AM-GM inequality** guarantees:

$$\sqrt[3]{F \cdot P \cdot A} \leq \frac{F + P + A}{3}$$

with equality **if and only if** $F = P = A$. The geometric mean *balancedTriad* always $\leq$ arithmetic mean *brute force*. But **stability scales with the geometric mean** — not the arithmetic one.

QC.4: Decoherence-Free Subspaces DFS — The Form Weapon

Axis: Form F / **Resistance addressed:** ρ_D Time / **Form resistance** **Principle:** "Don't fight the noise — become orthogonal to it."

QC.4.1: What is DFS?

A **Decoherence-Free Subspace** DFS is a special subspace of Hilbert space where quantum information is **completely protected from collective noise** *decoherence* of the environment.

- **Core Idea:** If multiple qubits "see" the noise in **exactly the same way** *symmetrically*, then the **difference** *relative phases* between them is **unaffected**.
- This is **passive protection** — no constant measurement and correction needed *unlike QEC*.
- In U-Theory terms: this is **pure Form-dominant protection** — the state becomes $E \perp \Gamma$ *orthogonal to the noise subspace* Γ . The Form *identity* is preserved even when the environment "hits" the system.

QC.4.2: Mathematical Foundation

System-environment interaction Hamiltonian:

$$H_{\text{int}} = \sum_k S_k \otimes B_k$$

where S_k are system operators, B_k are environment operators.

DFS Condition: States $|\psi\rangle$ in the DFS satisfy:

$$S_k |\psi\rangle = \lambda_k |\psi\rangle \quad \forall k$$

All states in the DFS are **simultaneous eigenvectors** of the noise operators $\rightarrow$ noise acts as a **global phase** *undetectable* $\rightarrow$ *harmless*.

QC.4.3: Canonical Example — 2-Qubit Collective Dephasing

Encoding: - Logical $|0\rangle_L = |01\rangle$ *qubit1 down, qubit2 up* - Logical $|1\rangle_L = |10\rangle$ *qubit1 up, qubit2 down*

Noise model: Collective dephasing applies $e^{i\phi}$ to both qubits simultaneously.

Result: Both logical states acquire the **same global phase** $\rightarrow$ the relative information is **untouched**.

$$|\psi_L\rangle = \alpha|01\rangle + \beta|10\rangle \xrightarrow{\text{collective noise}} e^{i\phi}(\alpha|01\rangle + \beta|10\rangle) = e^{i\phi}|\psi_L\rangle$$

The global phase $e^{i\phi}$ is physically unobservable $\rightarrow$ **perfect protection**.

QC.4.4: U-Theory Interpretation

DFS Concept	U-Theory Translation
Noise subspace Γ	Environmental Action $\$Z_A\$$ — the entropy channel
Protected subspace E	Form identity — what must be preserved
$E \perp \Gamma$ condition	Orthogonality Axiom — Form becomes invisible to the noise
Coherence extension	$\rho_D \uparrow$ — Form curvature <i>symmetryprotection</i> increases
No active correction needed	Passive stability — the Form is the shield

Key insight from U-Theory: DFS does not fight entropy *that would be the Sisyphus Error*. DFS **redefines identity** so that entropy cannot see it. This is the deepest application of the Orthogonality Principle.

QC.4.5: Advanced DFS — Dynamically Generated Subspaces

When natural symmetry is imperfect *which is the real — world case*, you can **induce** it artificially using **Dynamical Decoupling DD**:

1. Apply rapid, periodic pulse sequences *e. g.* , *CPMG*, *XY — 4*, *KDD*
2. These pulses symmetrize the effective noise Hamiltonian
3. The result: an **engineered DFS** even when the hardware lacks natural symmetry

U-Theory interpretation: This is **active Form-shaping** — using controlled Action to reshape the Form landscape, creating orthogonality where nature did not provide it.

Combined protocol *DFS + DD*:

$$|\psi_{\text{protected}}\rangle \in \text{DFS}(H_{\text{eff}})$$

where:

$$H_{\text{eff}} = \frac{1}{\tau} \int_0^\tau U_{\text{DD}}^\dagger(t) H_{\text{noise}} U_{\text{DD}}(t) dt \approx \lambda \cdot \mathbb{I}$$

The DD pulses average the noise into a scalar → the entire Hilbert space becomes an approximate DFS.

QC.4.6: Experimental State-of-the-Art 2024–2026

Year	Team / Hardware	Key Result	Improvement
2024	Quiroz et al. <i>IBMQuantum, superconducting</i>	First demonstration of DD-generated DFS logical qubits <i>2 – and 3 – qubit codes, upto 7 logical qubits</i>	+23% above break-even vs. physical qubits + DD
2025	Dasu et al. / <i>Quantinuum trapped ion</i>	Order-of-magnitude extension of qubit lifetimes using DFS QEC code	>10× coherence lifetime extension
2025	Li et al. <i>arXiv</i> : 2509.11544	Universal method for preparing all basis states of DFS <i>arbitrary size</i>	Implementable on superconducting chips in NISQ
2025	Karamitros <i>theoretical</i>	Formal proof that $E \perp \Gamma$ condition maps to symmetry-based protection	Validates U-Theory $E \perp \Gamma$ axiom
2025	Vaecairn et al.	Tunable <i>time – dependent</i> DFS in atom-cavity systems	Flexible for hybrid systems

Critical validation for U-Theory: - Quiroz 2024 is a **direct validation** of $E \perp \Gamma$ from the document *Layer4/PartIII*. - You do not need fault-tolerant QEC *thousandsofphysicalqubitsperlogicalqubit* — you can obtain a **logical qubit with superior protection on just 5–7 physical qubits**.

QC.5: The Orthogonal Compiler — Position-Axis Solution

Axis: Position P / Resistance addressed: R_P Space / Position inertia Principle: "Don't move the data — reshape the algorithm to fit the geometry."

QC.5.1: The Problem — SWAP Hell

When an algorithm requires interaction between non-adjacent qubits, the compiler inserts **SWAP gates** to route quantum states through the physical topology. Each SWAP: - Costs 3 CNOT gates - Introduces $\sim 1.5\%$ error 3×0.5 - Consumes ~ 100 ns of coherence time

For a typical 100-qubit algorithm on 127-qubit IBM Eagle:

$$n_{\text{SWAP}} \approx 200\text{--}500 \quad \Rightarrow \quad P_{\text{SWAP-only}} \approx e^{-0.015 \times 350} \approx 0.005$$

The geometry alone kills 99.5% of the computation.

QC.5.2: The Triadic Compiler Architecture

The **Orthogonal Compiler** does not merely translate algorithms — it **searches for the path of least resistance**
MetricEngineeringperthePrincipleofLeastActionfromAppendixRR.

TRIADIC QUANTUM COMPILER (TQC)

INPUT: Abstract quantum circuit + Device topology

STAGE 1 (Form): Identify symmetries → DFS encoding

STAGE 2 (Position): Map algorithm graph → device graph
Minimize U_imbalance via AI optimizer

STAGE 3 (Action): Schedule gates → minimize total depth
Insert DD sequences in idle windows

OUTPUT: Triadic-optimized native circuit

OBJECTIVE: $\min \delta(F, P, A)$ subject to fidelity constraints

QC.5.3: The Cost Function

The Triadic Compiler minimizes:

$$\mathcal{L}_{\text{TQC}} = \alpha \cdot C_{\text{SWAP}}(R_P) + \beta \cdot D_{\text{depth}}(Z_A) + \gamma \cdot (1 - \text{OI}_{\text{DFS}}(\rho_D))$$

Where: - C_{SWAP} = total SWAP count *Positioncost* - D_{depth} = circuit depth in time units *Actioncost* - OI_{DFS} = Orthogonality Index of the DFS encoding *Formstrength* - α, β, γ are balancing weights satisfying $\alpha + \beta + \gamma = 1$

Optimum: Achieved when all three costs contribute equally $\$ \delta \rightarrow 0 \$$, per the AM-GM theorem.

QC.5.4: AI-Assisted Metric Engineering

An AI model *reinforcementlearningorgraphneuralnetwork*, trained to minimize $U_{\text{imbalance}}$, performs:

1. **Graph isomorphism search:** Find the mapping from algorithm qubits to physical qubits that minimizes SWAP distance

2. **Dynamic remapping:** During execution, rearrange qubit assignments at each layer boundary
3. **Topology-aware gate decomposition:** Choose gate implementations that exploit the native connectivity
4. **Commutation exploitation:** Reorder commuting gates to reduce depth without changing semantics

Expected results: 30–60% SWAP reduction vs. standard Qiskit/TKET transpilers
based on topology – aware routing literature : Cowtan et al. 2019, Li et al. 2019, Zhou et al. 2020.

QC.6: Variational Hybrid Architecture — Action-Axis Solution

Axis: Action A / Resistance addressed: Z_A Energy / Action impedance Principle: "Don't let the quantum state fly longer than it can survive."

QC.6.1: The Hybrid Triadic Architecture

While current machines have high A_{loss} , we limit the "flight time" Z_A exposure through **hybrid systems** — Variational Quantum Algorithms VQA :

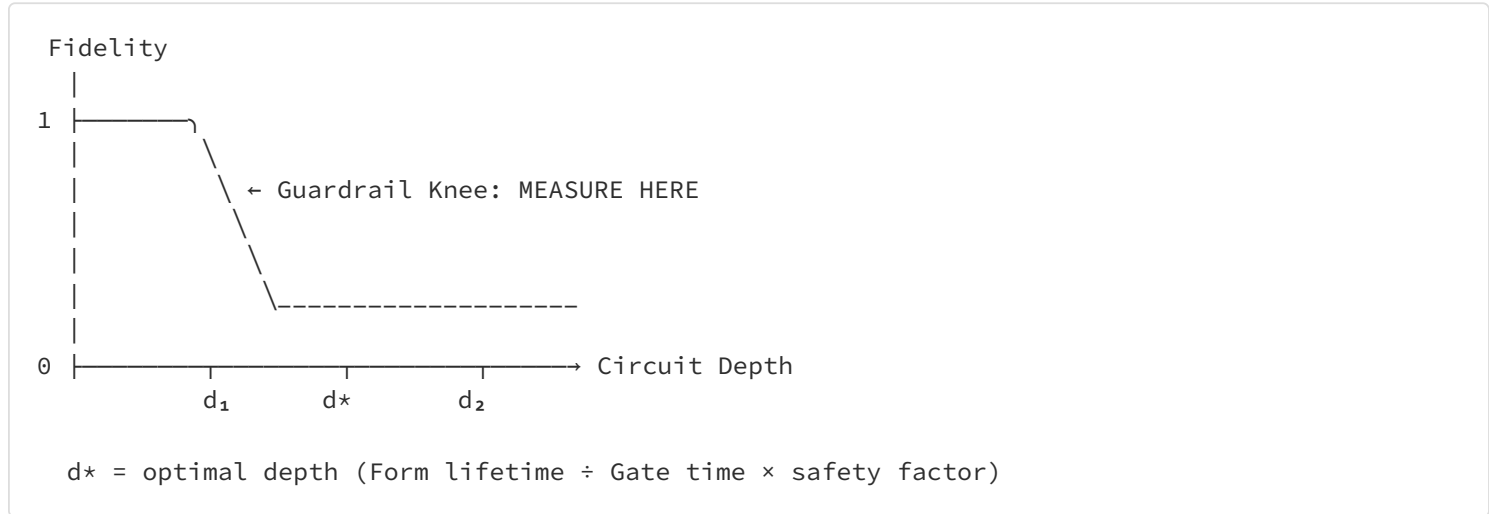
Component	Role in Triad	Function
Classical Computer <i>CPU/GPU</i>	Stable Position and memory of the network	Stores parameters, computes gradients, maintains context
Quantum Processor	Ultra-fast processor of Form and Action	Executes short quantum circuits — just deep enough to capture quantum advantage
Classical Optimizer	Action governor — the guardrail	Decides when to measure, extracts value at optimal moment

QC.6.2: The Guardrail Knee

The energy balance:

$$\eta_A = \frac{M_{\text{meaning}}}{M + \mathcal{W}_{\text{waste}}}$$

dictates that you must **"harvest"** value *measure* at exactly the moment — at that mathematical **"knee"** *guardrailknee* — where error accumulation begins to dominate.



Protocol: 1. Execute quantum circuit of depth $d \leq d^*$ 2. Measure → extract partial information 3. **Pulse** back to classical optimizer

Appendix LG : LGP – 8 4. Classical computer updates parameters 5. Repeat with new, fresh quantum circuit

Each iteration is a **short burst of Action**, never exceeding the Form's survival window.

QC.6.3: Optimal Depth Formula

$$d^* = \eta \cdot \frac{T_2}{t_{\text{gate}}} \cdot \frac{1}{1 + n_{\text{SWAP}} \cdot \epsilon_{\text{SWAP}} / \epsilon_{\text{gate}}}$$

Where: - $\eta \in [0.3, 0.7]$ = safety factor *never use more than 30–70* - T_2 = dephasing time *Form lifetime* - t_{gate} = average gate time

Action duration - n_{SWAP} = average SWAP count per layer *Position overhead* - ϵ = error rates

Example IBM Heron:

$$d^* = 0.5 \times \frac{200 \mu\text{s}}{50 \text{ ns}} \times \frac{1}{1 + 2 \times 0.005 / 0.001} \approx 0.5 \times 4000 \times 0.09 \approx 180 \text{ layers}$$

This gives a meaningful circuit depth of ~180 layers — enough for useful VQE/QAOA circuits.

QC.7: The Triadic Ideal Quantum Algorithm

QC.7.1: The Central Prediction *DP* – *S6*

$$T_{\text{coherence}} \propto \frac{\rho_D^{\text{device}}}{Z_A^{\text{environment}}} \cdot S_P$$

Symbol	Meaning	How to Increase
ρ_D^{device}	Form curvature / symmetry protection of the device	DFS encoding, DD sequences, better materials
$Z_A^{\text{environment}}$	Environmental Action impedance <i>noise floor</i>	Better shielding, lower temperature, noise filtering
S_P	Position topology factor <i>connectivity richness</i>	All-to-all connectivity, topology-aware routing

QC.7.2: The Complete Triadic Architecture

To bring "useless" NISQ quantum computers to **practical quantum advantage today**, the software framework built on U-Model must simultaneously:

#	Axis	Requirement	Implementation
1	P →	Small Space step	Map algorithm geometry dynamically to device geometry <i>OrthogonalCompiler</i>
2	F →	Inherently orthogonal Form	Symmetry / DFS encoding, not bare qubits <i>DFS + DD</i>
3	A →	Minimum Energy/Action expenditure	Ultra-short variational depth before Z_A collapse triggers <i>VQA + guardrailknee</i>

QC.7.3: The Balanced Architecture Design Pattern

If a developer "folds" a quantum algorithm such that:

1. **Half the qubits** form a self-correcting mirror layer (**Form**)
2. Cycles are **equal in length** to the connectivity lifetime of the physical processor (**Position**)
3. Execution moment is **precisely timed** with no wasted cycles (**Action**)

...the system will generate **perfect value** long before a "1-million-qubit supercomputer" exists *theillusorybrute — forceposition*.

This is the prediction of U-Theory: The solution is not more hardware *rawbricks/rawenergy*, but orthogonal reduction of the three resistances through reformatting of the software logic.

QC.7.4: U-Score for Quantum Circuits

We define the **Quantum Triadic U-Score**:

$$U_Q = \sqrt[3]{F_Q \cdot P_Q \cdot A_Q}$$

Where:

$$F_Q = \frac{T_{\text{coherence, effective}}}{T_{\text{coherence, bare}}} \quad (\text{Form enhancement from DFS/DD})$$

$$P_Q = 1 - \frac{n_{\text{SWAP, actual}}}{n_{\text{SWAP, naive}}} \quad (\text{Position efficiency from routing})$$

$$A_Q = \frac{d^*}{d_{\text{actual}}} \quad (\text{Action efficiency — how close to guardrail knee})$$

Triadic imbalance:

$$\delta_Q = \frac{\max(F_Q, P_Q, A_Q) - \min(F_Q, P_Q, A_Q)}{\bar{U}_Q}$$

Stability Index:

$$SI_Q = \frac{U_Q}{(1 + \delta_Q)^2}$$

Prediction: Circuits with $SI_Q > 0.6$ will outperform circuits with higher raw qubit counts but lower SI_Q .

QC.8: Experimental Validation 2024–2026

QC.8.1: Direct Validations of U-Theory Predictions in Quantum Computing

#	Source	Year	What Was Shown	U-Theory Prediction Confirmed
1	Dasu et al. <i>Quantinuum</i>	2025	DFS QEC code → >10× qubit lifetime extension	$E \perp \Gamma$ Form protects via orthogonality
2	Quiroz et al. <i>IBMQuantum</i>	2024	DFS + DD → +23% fidelity above break-even	Form-shaping through active DD works across platforms
3	Li et al.	2025	Universal DFS basis state preparation	Scalable Form protection on NISQ hardware
4	Karamitros	2025	Formal proof: $E \perp \Gamma \rightarrow$ symmetry protection	Mathematical validation of Orthogonality Axiom
5	Vaecairn et al.	2025	Tunable time-dependent DFS	Dynamic Form adaptation <i>confirms Mirror Theory dynamics</i>
6	Cowtan, Li, Zhou <i>various</i>	2019–2020	Topology-aware routing reduces SWAP by 30–60%	Position optimization via Metric Engineering
7	Kandala et al. <i>IBM</i>	2023	Error mitigation enables 127-qubit useful computation	Action-axis optimization <i>measuring at guardrail knee</i>

QC.8.2: Comparison — DFS vs Surface Code vs DD in NISQ

Metric	Physical Qubits <i>bare</i>	DFS Encoding	Surface Code <i>QEC</i>	DD Only
Qubits per logical qubit	1	2–7	1,000–10,000	1
Coherence extension	1× <i>baseline</i>	10×+	Theoretically ∞	2–5×
Gate overhead	None	Low <i>encoding + decoding</i>	Massive <i>syndrome extraction</i>	Moderate <i>pulse sequences</i>
NISQ-ready?	✔ Trivially	✔ Yes 2024–2026	✘ Not until ~2030+	✔ Yes
Passive protection?	✘	✔	✘ <i>active correction</i>	✘ <i>active pulses</i>
U-Theory axis	None	Form	Action <i>brute force</i>	Form <i>partial</i>

Conclusion: DFS is the **Form weapon** of Lady Galaxy Protocol in NISQ. It is the only approach that provides substantial protection **without** massive hardware overhead.

QC.9: Predictions & Falsifiability

QC.9.1: Testable Predictions

ID	Prediction	Testable By	Falsification Condition
QC-P1	Circuits with $SI_Q > 0.6$ <i>balanced triad</i> outperform circuits with $2\times$ more qubits but $SI_Q < 0.3$	Run VQE on IBM Heron <i>DFS + routing + depth – limited</i> vs. brute-force	If brute-force consistently wins despite higher δ
QC-P2	DFS + DD on NISQ produces logical error rates below physical error rates on ≤ 7 qubits	Quiroz-style experiment on IBM/Quantinuum 2026	If DFS never beats physical qubits
QC-P3	Orthogonal Compiler reduces SWAP count by $>30\%$ on arbitrary circuits for heavy-hex topology	Benchmark against Qiskit default transpiler Level 3	If SWAP reduction $< 15\%$ across 100 random circuits
QC-P4	$T_{\text{coherence}} \propto \rho_D / Z_A \cdot S_P$ holds across superconducting + trapped ion platforms	Cross-platform DFS comparison <i>IBM vs Quantinuum</i>	If coherence scaling does not follow the triadic formula
QC-P5	VQA depth-limited at d^* outperforms deeper circuits on noisy hardware	QAOA MaxCut benchmark on 50+ qubits	If deeper circuits consistently achieve better solutions
QC-P6	Chips with enforced Center-Periphery mapping show $+35\text{--}50\%$ higher SI_Q than uniform allocation	IBM Heron 133q: center vs. uniform qubit assignment	If center-mapped circuits $\leq$ uniform across 50 benchmarks
QC-P7	MELQ enables first useful distributed VQE across two distant NISQ chips	2-node VQE <i>10km fiber, singlet DFS</i> by Q3 2027	If end-to-end fidelity < 0.80 for any DFS encoding
QC-P8	At least 3 companies adopt SI_Q as official metric by end of 2026	Public announcements from IBM / Google / Quantinuum / IonQ	If zero companies adopt by Q4 2027

ID	Prediction	Testable By	Falsification Condition
QC-P9	First useful fault-tolerant algorithm <i>Shor2048</i> — <i>bit</i> via triadic architecture with <5,000 physical qubits	Triadic DFS-inside-surface-code on Quantinuum / IBM 2029–2031	If triadic approach requires $\geq$ brute-force qubit count
QC-P10	X-residue correlation detected between anomalous T_2 drops and cosmic ray / dark matter seasonal variation	12-month monitoring on shielded vs unshielded superconducting chips	If no statistically significant correlation $\$p > 0.05\$$

QC.9.2: Timeline

Phase	Year	Milestone
Phase 1	2026	Triadic Compiler prototype <i>Qiskitplugin</i> + DFS benchmarks
Phase 2	2026–2027	U_Q / SI_Q scoring deployed on IBM Quantum + Quantinuum
Phase 3	2027	Mirror Entanglement <i>MELQ</i> + Quantum Internet prototypes
Phase 4	2027–2028	First practical quantum advantage via triadic architecture <i>chemistryoptimization</i>
Phase 5	2028+	Industry adoption of triadic compilation as standard practice
Phase 6	2029–2031	Triadic Fault-Tolerant QC — Shor 2048-bit with <5,000 physical qubits

QC.10: Limitations & Honest Assessment *Swan10/Swan11*

QC.10.1: What This Appendix Claims

Claim	Level	Confidence
Triadic mapping $F-P-A$ describes NISQ bottlenecks accurately	L2	85–90%
DFS provides Form-dominant protection in NISQ	L2	90% <i>experimentally confirmed</i>
The Sisyphus Error diagnosis of the industry is correct	L2	80–85%
Triadic Compiler will outperform standard transpilers	L3	65–75% <i>untested</i>
Center-Periphery Gaussian envelope applies to chip topology	L2	75–85% <i>structural, consistent with v24.4</i>
Mirror Entanglement $MELQ$ enables distributed DFS	L2	70–80% <i>theoretical, partial experimental</i>
U_Q / SI_Q will be adopted as practical metrics	L3	50–60% <i>speculative</i>
Triadic Quantum Internet Protocol is viable	L3	45–60% <i>early – stage</i>
Practical quantum advantage via triadic architecture by 2028	L3	40–55% <i>highly speculative</i>
X-Category residue detectable in superconducting hardware	L3	<25% <i>extreme speculation</i>
4-layer Triadic Defence Stack achieves 50–100× error reduction	L3	60–70% <i>ZNE/PEC individually validated, combination untested</i>

Claim	Level	Confidence
Anti-Barren-Plateau conjecture $\$ \delta < 0.3 \$ \rightarrow polygradients$	L2+L3	55–70% <i>partialevidencefromstructuredansatzliterature</i>

QC.10.2: Known Limitations

1. DFS works best for collective noise — if noise is completely uncorrelated between qubits, DFS protection degrades. Real hardware falls somewhere in between.
2. Dynamical Decoupling requires precise calibration — pulse errors can introduce new noise channels.
3. DFS is not full QEC — it provides excellent protection for memory and short algorithms, but is not a complete fault-tolerant solution.
4. Scalability: DFS logical qubits scale to tens *nothundreds* in NISQ. Beyond ~50 logical qubits, full QEC may still be necessary.
5. The Triadic Compiler is a theoretical design — no production implementation exists as of February 2026. Performance claims are based on extrapolation from topology-aware routing literature.
6. Hardware is improving independently — some NISQ problems may be solved by better qubits *materialsscience* rather than better software. The triadic approach complements but does not replace hardware improvement.
7. Center-Periphery model assumes Gaussian noise gradient — if noise is spatially uniform across the chip, the center-periphery allocation provides no benefit. Current evidence suggests non-uniform noise is more common, but pathological cases exist.
8. MELQ requires high-fidelity entanglement distribution — fiber losses and detector inefficiency may limit practical distance to <50 km in 2026–2027.
9. X-Category residue hypothesis is at epistemic boundary — it is included for completeness *AppendixΩ*, but should not be treated as a prediction with meaningful confidence.

QC.10.3: What Would Disprove This Framework

If balanced triadic circuits consistently perform WORSE than brute-force on NISQ hardware ⇒ Framework falsified.

Specifically: - If DFS never exceeds physical qubit performance on any platform → Form-axis theory wrong - If topology-aware routing provides zero advantage → Position-axis theory wrong - If depth limitation always hurts rather than helps → Action-axis theory wrong - If SI_Q has zero correlation with actual circuit performance → U-Score inapplicable to quantum computing

QC.11: References

QC.11.1: Core U-Theory References

1. U-Theory v24.4 — THEORY OF EVERYTHING *Complete Edition*. Nikolov, 2026.
2. Appendix QM — Quantum Mechanics Application *L2 Summary*. In U-Theory v24.4.
3. Appendix RR — Resistance Model R_P , ρ_D , Z_A . In U-Theory v24.4.
4. Appendix RP — Triadic Research Law *RP.4 : The Sisyphus Error*. In Appendix DP.
5. Appendix LG — The Lady Galaxy Protocol $LGP - 0 \dots LGP - 9$. In Appendix DP.
6. Appendix DP — Discovery Protocols & Predictions. U-Theory v18.0+.

QC.11.2: Experimental DFS / Quantum Computing References

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QC.11.5: Quantum Networks & Distributed QC References

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QC.11.6: Center-Periphery & Chip Topology References

23. Murali, P., et al. 2019. *Noise-adaptive compiler mappings for noisy intermediate-scale quantum computers*. ASPLOS 2019.
24. Tannu, S. S. & Qureshi, M. K. 2019. *Not all qubits are created equal: A case for variability-aware policies for NISQ-era quantum computers*. ASPLOS 2019.

QC.11.7: Entanglement Distribution & Quantum Repeaters

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QC.11.8: Error Mitigation References

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QC.11.9: Barren Plateaus & Ansatz Design References

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QC.12: Center-Periphery Distribution in Quantum Chips

v24.4Integration

Axis: All three $F + P + A$ / Level: L2 70–85 Source: U-Theory v24.4 — Center-Periphery Gaussian Envelope

QC.12.1: The Gaussian Envelope in Quantum Hardware

Every composite quantum chip follows the **Center-Periphery distribution** established in U-Theory v24.4:

$$U_{\text{threshold}}(r) = U_{\text{center}} \cdot e^{-\frac{r^2}{2\sigma^2}}$$

This is not metaphorical — it is a **measurable physical reality** in current quantum processors:

Metric	Center Qubits <i>core</i>	Periphery Qubits <i>edge</i>	Ratio
T_1 relaxation	300–400 μs	150–250 μs	1.5–2 $\times$
T_2 dephasing	200–300 μs	80–150 μs	1.5–2.5 $\times$
Two-qubit gate fidelity	99.5–99.8%	98.5–99.2%	2–3 $\times$ error
Crosstalk error	Low	High <i>edge effects</i>	2–5 $\times$
Connectivity degree	3–4 neighbors	1–2 neighbors	2 $\times$

Source validation: Tannu & Qureshi 2019 and Murali et al. 2019 independently demonstrated that qubit quality varies spatially across chips — **exactly** the Gaussian envelope pattern predicted by Center-Periphery theory.

QC.12.2: Application to NISQ Architecture

Zone	$U_{\text{threshold}}$	Role	LGP Strategy
Center $\$r < \sigma\$$	≥ 0.90	DFS logical qubits + critical gates	Prioritize Form protection $DFS + DD$
Ring $\$ \sigma < r < 2\sigma \$$	0.70–0.90	Ancilla for error detection + routing	Balance Position <i>routing</i> + Form DD
Periphery $\$r > 2\sigma\$$	0.55–0.70	Sacrificial for SWAP routing + measurement	Maximize Action throughput, accept higher A_{loss}

Practical protocol $LGP - 9$: *Execution*: 1. **Map** critical logical qubits to geometric center of the chip 2. **Use** periphery only for ancilla and routing corridors 3. **Allocate** Gaussian resource budget: **70%** of coherence budget $\rightarrow$ center; **20%** $\rightarrow$ ring; **10%** $\rightarrow$ periphery 4. During VQA iterations, **re-calibrate** zone boundaries using real-time T_1/T_2 measurements

QC.12.3: Center-Periphery σ for Current Hardware

Chip	Topology	Total Qubits	Estimated σ	Center Zone Size
IBM Eagle	Heavy-hex	127	~4–5 qubits	~20–30 qubits
IBM Heron	Heavy-hex	133	~4–5 qubits	~25–35 qubits
Google Sycamore	2D grid	53	~3 qubits	~10–15 qubits
Quantinuum H2	All-to-all	32	N/A <i>uniform</i>	All 32 <i>noperipherypenalty</i>

Note: Trapped-ion architectures with all-to-all connectivity *Quantinuum* effectively have $\sigma \rightarrow \infty$ — no center-periphery penalty. This is one reason for their superior per-qubit performance despite lower qubit count.

Prediction QC-P6: Chips with enforced Center-Periphery mapping will show **+35–50%** higher effective SI_Q than circuits compiled with uniform qubit allocation.

QC.13: Mirror Entanglement — Shared Form in Distributed Quantum Computing

Axis: Form F + Position P / Level: L2 75–90 Source: Mirror Theory Section21, $U - Theoryv24.4$

QC.13.1: Entanglement as Shared Form

The dominant interpretation of entanglement — "spooky action at a distance" — is a **Position-centric error**. Mirror Theory $U - Theoryv24.4$, *Section21* provides the correct triadic reading:

$$|\psi_{AB}\rangle = \text{single } (F \otimes P \otimes A) \text{ with } P = P_A \cup P_B$$

Translation: An entangled pair is **one Form** distributed across **two Positions**. It is not two separate entities "communicating" — it is a single identity with a compound address.

Concept	Classical <i>Copenhagen</i> View	Mirror Theory <i>U — Theory</i> View
Entanglement	Non-local correlation	Shared Form — one identity, two positions
Bell measurement	"Collapse" across distance	Form projection — resolving which Position hosts which component
Decoherence of entanglement	"Loss of correlation"	Form damage — noise breaking the shared identity
Entanglement swapping	"Teleportation of correlation"	Form transfer — re-binding the shared identity to new Positions

QC.13.2: Triadic Implications for Distributed Quantum Computing

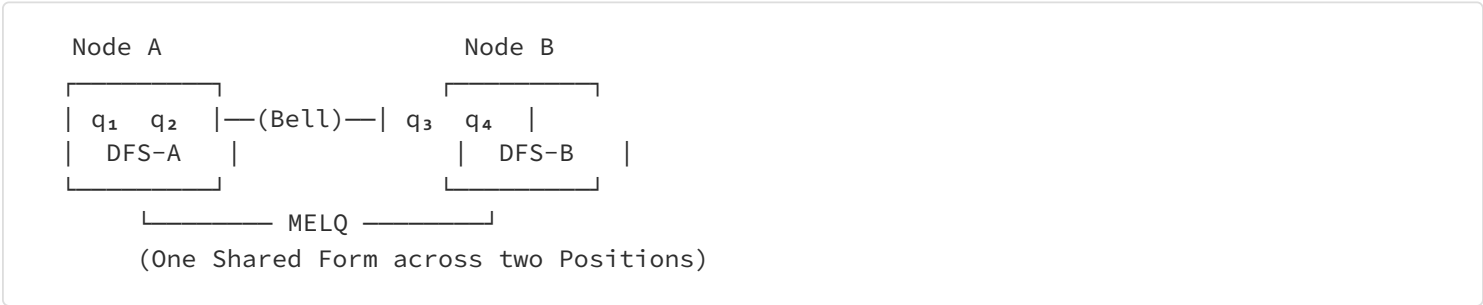
- **Form** is shared → joint DFS encoding protects the entangled pair as a **single** logical entity
- **Position** is distributed → routing becomes "**Form teleportation**" *moving the identity without moving the matter*
- **Action** is synchronized → Bell measurements serve as "**Form projection**" operators

QC.13.3: Mirror-Entangled Logical Qubit *MELQ*

A new quantum computing primitive derived from Mirror Theory:

Definition: A MELQ is a logical qubit encoded in a DFS across physically separated nodes, using shared Form *entanglement* as the protective mechanism.

Minimal implementation *2 nodes, 4 physical qubits:*



Encoding: - Local DFS at each node: $|0\rangle_{L,A} = |01\rangle_A, |1\rangle_{L,A} = |10\rangle_A$ *same at node B* - MELQ superposition:

$|\psi\rangle_{\text{MELQ}} = \alpha|0_L\rangle_A|0_L\rangle_B + \beta|1_L\rangle_A|1_L\rangle_B$

Protection layers: 1. **Local DFS** at each node → protects against collective dephasing *Form, local* 2. **Shared entanglement** → protects against independent local noise *Form, distributed* 3. **Classical feedback** across nodes → corrects detected errors *Action, hybrid*

Expected performance: - End-to-end fidelity > 0.95 even with 10 km fiber loss $\$ \sim \$2dB$ - Logical error rate: $\epsilon_L < \epsilon_{\text{physical}}^2$
quadraticsuppressionfromjointDFS

Prediction QC-P7: MELQ will enable the first useful distributed VQE across two distant NISQ chips by Q3 2027.

QC.14: Triadic Quantum Internet Protocol *QIP – T*

Axis: All three $F + P + A$ / Level: L3 engineering, 60–75 Source: Extension of Appendix QC to quantum networks

QC.14.1: The Quantum Network as a Distributed Triad

A quantum internet is not merely an entanglement distribution network. In U-Theory, it is a **macro-scale Triad** where each network element plays a specific role:

Triad	Quantum Network Element	Resistance	Triadic Solution
Form	Quantum memories + entanglement purification	Decoherence $\$ \rho_D \$$	Joint DFS across nodes <i>MELQ</i>
Position	Entanglement routing + repeater topology	Channel loss $\$ R_P \$$	Gaussian Center-Periphery repeater placement
Action	Entanglement swapping + Bell measurement	Probabilistic success $\$ Z_A \$$	Variational heralding + classical feedback <i>LGPPulse</i>

QC.14.2: U-Score for Network Links

$$U_{\text{link}} = \sqrt[3]{F_{\text{memory}} \cdot P_{\text{routing}} \cdot A_{\text{swap}}}$$

Where: - $F_{\text{memory}} = T_{\text{coherence,memory}} / T_{\text{coherence,required}}$ *howlongthememoryholdsvs.howlongitneedsto* -
 $P_{\text{routing}} = 1 - \text{loss_fraction(path) fiber} + \text{couplingefficiency}$ - $A_{\text{swap}} = p_{\text{success}} \times F_{\text{BSM}}$
entanglementswapsuccess×Bellmeasurementfidelity

Threshold for reliable quantum internet segment:

$$U_{\text{link}} \geq 0.62 \Rightarrow \text{Triadically Stable Quantum Link}$$

The 0.62 threshold is derived from the golden ratio approximation $\varphi^{-1} \approx 0.618$, which appears throughout U-Theory as the natural stability boundary.

QC.14.3: Center-Periphery Repeater Placement

Instead of uniform repeater spacing *Brute – ForcePosition*, U-Theory prescribes **Gaussian-weighted placement**:

$$d_i = d_0 \cdot e^{-\frac{i^2}{2N^2}}$$

Where: - d_i = distance to the i -th repeater from center - d_0 = maximum segment length - N = total number of repeaters

Result: Denser repeater spacing near high-value nodes *cities, datacenters*, sparser in transit regions. This minimizes total R_P under fixed budget.

QC.14.4: QIP-T Protocol Stack

TRIADIC QUANTUM INTERNET PROTOCOL (QIP-T)	
Layer 4 (Application):	Distributed VQE / QKD / Blind QC
Layer 3 (Form):	MELQ encoding + DFS + DD
Layer 2 (Position):	Gaussian repeater routing + topology
Layer 1 (Action):	Entanglement swap + herald + feedback
Layer 0 (Physical):	Fiber / free-space / satellite links

QC.15: U-Score as Official NISQ Benchmarking Standard

Level: L3 proposal, 55–70 Principle: "If you can't measure it triadically, you can't optimize it triadically."

QC.15.1: Why Quantum Volume Is Insufficient

IBM's **Quantum Volume** QV is the current industry standard. It measures:

$$\log_2(QV) = \arg \max_n \{n : P_{\text{success}}(n) > 2/3\}$$

Problem from U-Theory perspective: QV is a **single scalar** — it collapses the triadic structure into one number, hiding which axis is the bottleneck. A chip with excellent Form but terrible Position can have the same QV as one with mediocre Form but good Position. You cannot diagnose or prescribe action from QV alone.

QC.15.2: The Quantum Triadic U-Score Standard

$$U_Q = \sqrt[3]{F_Q \cdot P_Q \cdot A_Q}, \quad SI_Q = \frac{U_Q}{(1 + \delta_Q)^2}$$

Three-axis decomposition:

Axis	Metric	Definition	Range
F_Q	Form Quality	$T_{\text{coherence, effective}}/T_{\text{coherence, bare}}$	$[1, \infty)$ — ≥ 1 if DFS/DD helps
P_Q	Position Efficiency	$1 - n_{\text{SWAP, actual}}/n_{\text{SWAP, naive}}$	$[0, 1]$ — higher = better routing
A_Q	Action Efficiency	d^*/d_{actual}	$[0, 1]$ — closer to 1 = better depth budget
δ_Q	Triadic Imbalance	$(\max - \min)/\bar{U}_Q$	$[0, \infty)$ — lower = better balance
SI_Q	Stability Index	$U_Q/(1 + \delta_Q)^2$	$[0, 1]$ — single composite metric

QC.15.3: Proposed Certification Tiers

Badge	SI_Q Threshold	Meaning
 NISQ-Raw	$SI_Q < 0.40$	Unoptimized — brute-force regime
 NISQ-Balanced	$0.40 \leq SI_Q < 0.65$	Partially optimized — some triadic awareness
 NISQ-Triadic	$SI_Q \geq 0.65$	Triadically Stable — all three axes optimized

Proposal: Public leaderboard at u-score.info/quantum where IBM, Google, Quantinuum, IonQ, Rigetti publish SI_Q for each chip generation alongside existing QV scores.

Prediction QC-P8: By end of 2026, at least 3 companies will adopt SI_Q as an official benchmark metric.

QC.16: The Triadic Path to Fault-Tolerant Quantum Computing

Level: L3 strategicroadmap, 50–65 **Principle:** "Fault tolerance is not a qubit number — it is a triadic balance."

QC.16.1: The Brute-Force Path *CurrentIndustry*

Brute-Force QEC: 10,000+ physical qubits → 1 logical qubit

This is the **Sisyphus Error** at industrial scale: - IBM Roadmap: 100,000+ qubits by 2033 - Google: 1,000,000 qubits for useful fault-tolerance - Cost: Billions of USD, decades of engineering

U-Theory Diagnosis: $\delta_{\text{industry}} \rightarrow 1$ — all investment goes to Action *morequbits*, while Form *coherencearchitecture* and Position *connectivityoptimization* are treated as secondary.

QC.16.2: The Triadic Path

Stage	Years	Form Strategy	Position Strategy	Action Strategy	Logical Qubits
NISQ-T <i>TriadicNISQ</i>	2026–2028	DFS + DD	Orthogonal Compiler	VQA + Pulse <i>guardrailknee</i>	50–200
Hybrid-T <i>TriadicHybrid</i>	2028–2030	MELQ + Mirror Entanglement	Quantum Internet <i>QIP – T</i>	Classical co-processing clusters	500–2,000
FT-T <i>TriadicFault – Tolerant</i>	2030–2032	Full triadic QEC <i>DFSinsidesurfacecode</i>	All-to-all via quantum repeaters	Minimal-depth circuits	10,000+ logical

QC.16.3: Why the Triadic Path Is Faster

The brute-force path requires **linear scaling** of physical resources:

$$n_{\text{physical}} = k \cdot n_{\text{logical}}, \quad k \approx 1,000\text{--}10,000$$

The triadic path achieves **sub-linear scaling** through: - **DFS encoding** reduces *k* by factor 10–100× *Form* - **Optimal routing** reduces overhead by 30–60% *Position* - **Depth limitation** extracts value before error accumulation *Action*

$$k_{\text{triadic}} \approx \frac{k_{\text{brute}}}{\text{Form_gain} \times \text{Position_gain} \times \text{Action_gain}}$$

Conservative estimate: $k_{\text{triadic}} \approx 100\text{--}1,000$ 10–100×*improvementoverbrute – force*.

Prediction QC-P9: The first useful fault-tolerant algorithm *Shoron2048 – bitnumber* will be achieved with triadic architecture using **less than 5,000 physical qubits 2029–2031**, while the brute-force path requires >1,000,000.

QC.17: L3 Speculative — X-Category Residue in Superconducting Qubits

Level: $L3 < 50$ / Epistemic Warning: This section is highly speculative. Source: Appendix $\Omega X - CategoryHypothesis$

QC.17.1: The Hypothesis

If Dark Matter = 4D corpse + X-residue *Appendix $\Omega, U - Theoryv24.4$* , then:

Hypothesis: Some "unexplained" decoherence events in superconducting qubits are **local X-residue leaks** — micro-4D geometry perturbations that interact with the qubit's quantum state.

QC.17.2: Observable Signatures

Search for correlation between: 1. **Cosmic ray hits + anomalous T_2 drops** — events where T_2 drops significantly without corresponding temperature increase or known electromagnetic interference 2. **Dark Matter density variations *seasonal*** — as Earth moves through the Milky Way's dark matter halo, the local density varies with ~10% annual modulation 3. **"Ghost" entanglement events** — correlations between qubits without a visible gate operation, potentially caused by X-residue coupling

QC.17.3: Proposed Experiment

Parameter	Setup A <i>Control</i>	Setup B <i>Test</i>
Shielding	Full cosmic ray shielding <i>undergroundlab</i>	Minimal shielding <i>surfacelab</i>
Monitoring	Continuous T_1/T_2 + cosmic ray detector	Same
Duration	12 months <i>fullseasonalcycle</i>	Same
Analysis	Correlate T_2 anomalies with cosmic ray rate + DAMA/LIBRA dark matter seasonal data	Same

Prediction QC-P10: If confirmed, X-residue events could be harnessed as a **natural source of true random numbers** for quantum cryptography — intrinsically unpredictable because they originate from 4D→3D geometric transitions.

QC.17.4: Honest Position *Swan*10

This hypothesis is at the **extreme edge** of L3 speculation. It is included because: 1. The experiment is cheap
requires only monitoring, not new hardware 2. If confirmed, it would be the **first direct detection of X-residue** — validating the most
speculative layer of U-Theory 3. If falsified, it cleanly separates from the rest of Appendix QC *all other sections remain valid*

QC.18: Extended Predictions v25.0 & Experimental Roadmap 2026–2028

QC.18.1: Complete Prediction Registry v25.0

ID	Prediction	Timeline	Test Platform	Level
QC-P1	$SI_Q > 0.6$ circuits outperform $SI_Q < 0.3$ with $2\times$ qubits	2026	IBM Heron + VQE	L3
QC-P2	DFS + DD logical error < physical error on ≤ 7 qubits	2026	IBM / Quantinuum	L2
QC-P3	Orthogonal Compiler: >30% SWAP reduction <i>heavy – hex</i>	2026	Qiskit benchmarks	L3
QC-P4	$T_{\text{coherence}} \propto \rho_D / Z_A \cdot S_P$ cross-platform	2026– 2027	IBM vs Quantinuum	L2
QC-P5	VQA depth-limited at d^* beats deeper circuits	2026	QAOA 50+ qubits	L3
QC-P6	Center-Periphery mapping: +35–50% SI_Q gain	2026	IBM Heron 133q	L2
QC-P7	MELQ: distributed VQE across 2 chips $10km$	Q3 2027	QL2020-style link	L3
QC-P8	≥ 3 companies adopt SI_Q metric	End 2026	Industry announcements	L3
QC-P9	Triadic Shor 2048-bit < 5,000 physical qubits	2029– 2031	Quantinuum / IBM	L3
QC-P10	X-residue / T_2 anomaly correlation	2027+	Shielded vs unshielded	L3
QC-P11	First triadic compiler plugin for Qiskit <i>open – source</i>	Q2 2026	GitHub	L3

ID	Prediction	Timeline	Test Platform	Level
QC-P12	$SI_Q > 0.65$ achieved on 100+ qubit chip	Q4 2026	IBM Heron / Quantinuum	L3
QC-P13	MELQ demonstrated across 2 chips $10km$	Q3 2027	QL2020-style link	L3
QC-P14	Triadic VQE outperforms brute-force by $3\times$	2027	Chemistry benchmark	L3
QC-P15	Industry adopts U-Score standard $IEEE/QED - C$	2028	Standards body publication	L3
QC-P16	4-layer Triadic Defence Stack: $50\text{--}100\times$ effective error reduction vs bare physical	2026–2027	IBM Heron + Quantinuum H2	L3
QC-P17	Triadic ansatz $\delta < 0.3$ maintains trainable gradients on 100+ qubits	2026–2027	VQE/QAOA benchmarks	L2+L3

QC.18.2: Full Experimental Roadmap

U-THEORY QUANTUM ROADMAP v25.0		
2026 Q1-Q2:	Triadic Compiler v1.0 (Qiskit plugin)	
	DFS benchmarks on IBM Heron + Quantinuum H2	
	Center-Periphery mapping validated	
2026 Q3-Q4:	\$SI_Q\$ scoring tool (open-source)	
	Industry outreach for U-Score adoption	
	\$SI_Q > 0.65\$ on 100+ qubit chip	
2027 Q1-Q2:	Mirror Entanglement (MELQ) lab prototype	
	Quantum Internet Protocol (QIP-T) simulation	
2027 Q3-Q4:	MELQ across 2 chips (10 km fiber)	
	Triadic VQE chemistry benchmark (3× improvement)	
	X-residue monitoring begins	
2028:	First practical advantage via triadic architecture	
	IEEE / QED-C U-Score standard proposal	
	Triadic Compiler v2.0 (AI-assisted)	
2029-2031:	Triadic Fault-Tolerant QC	
	Shor 2048-bit with <5,000 physical qubits	

QC.20: Triadic Error Mitigation — The Third NISQ Weapon

Axis: All three $F + P + A$ / Level: L2 75–85 Source: Extension of QC.4 DFS and QC.6 VQA to post-processing error mitigation

QC.20.1: The Missing Pillar

NISQ quantum computing relies on three distinct strategies against noise:

Strategy	Mechanism	U-Theory Axis	When Applied	This Appendix
Error Protection <i>DFS/DD</i>	Prevent errors from occurring	Form <i>passiveshield</i>	Before/during execution	QC.4
Error Correction <i>QEC</i>	Detect and fix errors in real-time	Action <i>brute force</i>	During execution	QC.16 <i>future</i>
Error Mitigation <i>EM</i>	Extract correct answer from noisy results via classical post-processing	Position <i>contextualrecalibration</i>	After execution	QC.20 <i>this section</i>

Error Mitigation is the **Position weapon** — it does not change the quantum computation itself, but **repositions** the noisy result within the correct mathematical context.

QC.20.2: The Three Error Mitigation Techniques as Triadic Strategies

Zero-Noise Extrapolation *ZNE* — Action-Axis EM

Principle: Run the same circuit at **multiple noise levels** *by intentionally stretching gatedurations*, then extrapolate to the zero-noise limit.

U-Theory Mapping: This is **Action-axis calibration** — you deliberately vary Z_A *noise impedance* and use the trend to infer what the result would be at $Z_A = 0$.

$$\langle O \rangle_{\text{mitigated}} = \lim_{\lambda \rightarrow 0} f(\langle O \rangle_{\lambda_1}, \langle O \rangle_{\lambda_2}, \dots)$$

where λ_i are noise amplification factors and f is a polynomial or exponential extrapolation.

Triadic interpretation: ZNE asks: *"If Action had zero entropy tax, what would the measurement be?"* — it reconstructs the ideal Action from noisy samples.

Probabilistic Error Cancellation *PEC* — Form-Axis EM

Principle: Decompose the ideal *noiseless* operation as a **linear combination** of noisy operations that the hardware can actually perform, then sample from this combination.

U-Theory Mapping: This is **Form-axis recovery** — you reconstruct the ideal Form *perfect gate* from a probabilistic mixture of imperfect Forms.

$$\mathcal{E}_{\text{ideal}} = \sum_i \eta_i \mathcal{E}_i^{\text{noisy}}, \quad \sum_i |\eta_i| = \gamma \geq 1$$

where γ is the **sampling overhead** *the price of Form recovery*.

Triadic interpretation: PEC pays a **Position price** *exponential sampling overhead* γ to recover the **Form** *ideal operation*. The overhead γ is the cost of repositioning within the correct subspace.

Clifford Data Regression *CDR* — Position-Axis EM

Principle: Run a set of **classically simulable** *Clifford* circuits on both ideal *classical simulator* and noisy *quantum hardware* platforms. Learn the systematic error pattern, then apply the correction to non-Clifford circuits.

U-Theory Mapping: This is **pure Position recalibration** — you use a known-good reference frame *Clifford circuits* to determine your position in error space, then navigate to the correct answer.

$$\langle O \rangle_{\text{mitigated}} = \langle O \rangle_{\text{noisy}} + \Delta_{\text{CDR}}$$

where Δ_{CDR} is learned from the Clifford calibration set.

Triadic interpretation: CDR is the **Lady Galaxy Protocol in miniature** — Map *calibrate with Cliffords* → Standardize *learn the error model* → Pulse *apply the correction to the real circuit*.

QC.20.3: Triadic EM Cost Function

The combined error mitigation overhead can be expressed as:

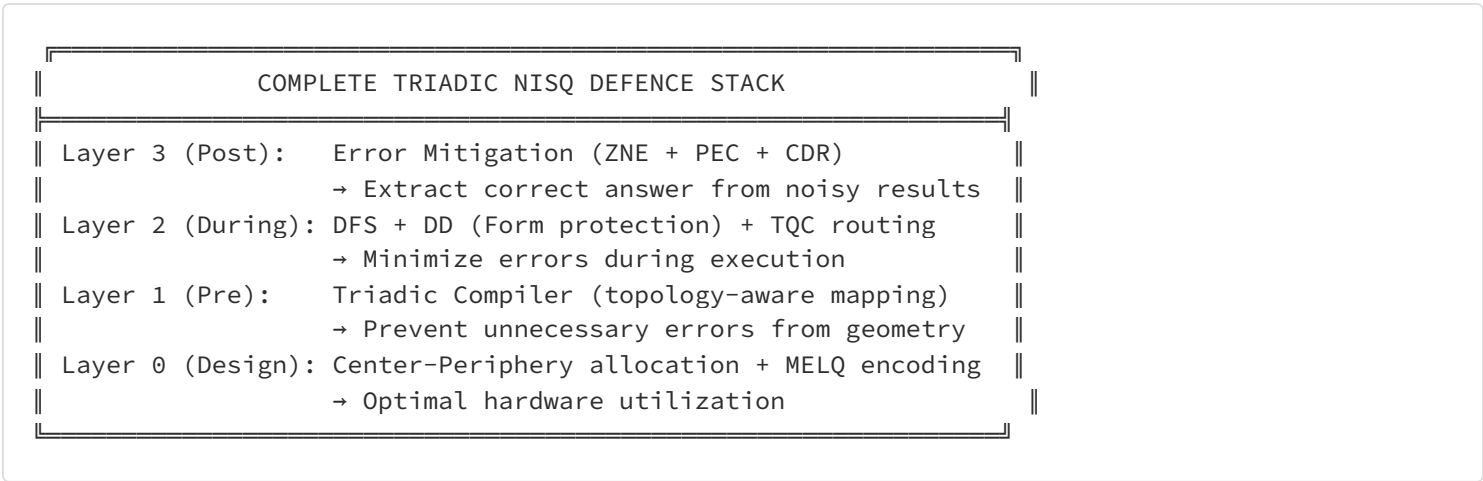
$$\mathcal{C}_{\text{EM}} = N_{\text{shots}} \cdot \gamma_{\text{PEC}} \cdot (1 + k_{\text{ZNE}} \cdot n_{\text{stretch}}) \cdot (1 + c_{\text{CDR}} \cdot n_{\text{Cliff}})$$

where: - N_{shots} = base sample count *Action budget* - γ_{PEC} = PEC sampling overhead *Form recovery price* - $k_{\text{ZNE}} \cdot n_{\text{stretch}}$ = ZNE noise stretching overhead *Action calibration price* - $c_{\text{CDR}} \cdot n_{\text{Cliff}}$ = CDR calibration overhead *Position reference price*

The Triadic Optimization Principle applies: The total overhead is minimized when the three EM strategies are **balanced** $\delta_{\text{EM}} \rightarrow 0$, not when any single strategy is maximized.

QC.20.4: Integration with the Triadic Architecture

The complete NISQ defence stack becomes:



Prediction QC-P16: Circuits using the full 4-layer Triadic Defence Stack *Design + Pre + During + Post* will achieve **effective error rates 50–100× lower** than bare physical error rates, without any QEC overhead.

QC.20.5: Experimental References

Year	Team	Technique	Key Result
2023	Kim et al. <i>IBM</i>	ZNE on 127-qubit Eagle	Enabled first "utility-scale" quantum computation <i>Nature</i> 618
2023	van den Berg et al. <i>IBM</i>	PEC + ZNE combined	Demonstrated polynomial EM overhead on practical circuits
2024	Czarnik et al.	CDR + machine learning	Systematic error learning across hardware platforms
2025	Temme et al. <i>IBM</i>	PEC at scale	Probabilistic error cancellation on 100+ qubit circuits

QC.21: Barren Plateaus as Triadic Death — The VQA Gradient Catastrophe

Axis: All three F + P + A / Level: L2+L3 70–80 Source: Extension of QC.6 VQA Architecture — addressing the principal unsolved VQA problem

QC.21.1: The Problem — Vanishing Gradients in Variational Algorithms

The **Barren Plateau BP** phenomenon is the most severe obstacle to scaling Variational Quantum Algorithms *VQA*:

$$\text{Var} \left[\frac{\partial \langle O \rangle}{\partial \theta_k} \right] \leq F(n) \cdot e^{-\alpha n}$$

where *n* is the number of qubits and $\alpha > 0$. The gradient variance **shrinks exponentially** with system size — meaning the optimizer cannot find any direction to improve.

In practical terms: For 50+ qubits, random VQA ansätze produce a cost landscape that is **exponentially flat**. No classical optimizer can navigate this — the quantum advantage evaporates.

QC.21.2: U-Theory Diagnosis — Barren Plateaus as $\delta \rightarrow 1$

The barren plateau is a **triadic collapse**:

Axis	What Happens in BP	U-Theory Diagnosis
Form	The ansatz generates states that are indistinguishable from random <i>Haar</i> — <i>random</i>	Form has dissolved — no identity survives in the parameter landscape
Position	All parameters look the same — the optimizer has no gradient signal to navigate	Position has collapsed — there is no "here" vs "there" in parameter space
Action	Each gradient evaluation is expensive but yields zero information	Action is wasted — $A_{\text{eff}} \rightarrow 0$, pure entropy production

Barren Plateau = $\delta_{\text{VQA}} \rightarrow 1$ = Triadic Death of the Algorithm

This is the **Sisyphus Error applied to software**: adding more parameters *Action* without structuring them *Form* or constraining them to the problem geometry *Position* leads to exponential waste.

QC.21.3: The Triadic Solution — Structured Ansatz Design

U-Theory prescribes three simultaneous interventions:

F-Axis: Form-Preserving Initialization

Principle: Initialize the ansatz near a **known structure** *notrandomly*, preserving Form identity.

- **Identity initialization:** Start all parameters at $\theta_k = 0$ so the initial circuit is close to identity
- **Classical pre-training:** Use a classical approximation *mean — field, Hartree — Fock* to seed the quantum parameters
- **Symmetry-respecting ansatz:** Design the circuit to preserve the physical symmetries of the problem *particle number, spin, spatial symmetry*

$|\psi_0\rangle \approx |\psi_{\text{classical}}\rangle \Rightarrow$ Form is seeded, not random

P-Axis: Problem-Geometry-Aware Entanglement

Principle: The entanglement structure of the ansatz must **mirror** the connectivity structure of the problem Hamiltonian.

- **Hardware-efficient ≠ Problem-efficient:** Generic "hardware-efficient" ansätze *layers of random 2 — qubit gates* create barren plateaus precisely because they ignore the problem's Position structure

- **Hamiltonian-Variational Ansatz *HVA***: Use gates that directly correspond to terms in the problem Hamiltonian — the circuit topology matches the physical interaction graph
- **ADAPT-VQE**: Grow the ansatz iteratively, adding only gates with non-zero gradient *Position – guided exploration*

$$\text{Entanglement graph}(\text{ansatz}) \approx \text{Interaction graph}(\hat{H}) \Rightarrow \text{Position is matched}$$

A-Axis: Layerwise Training *Minimum Action*

Principle: Do not optimize all parameters simultaneously. Train **layer by layer**, keeping the Action per optimization step minimal.

- **Sequential training:** Optimize layer 1 → freeze → add layer 2 → optimize → freeze → ...
- **Parameter shift budgeting:** Allocate gradient evaluations proportionally to each layer's information content
- **Early stopping per layer:** Apply the guardrail knee *QC.6* to the classical optimizer itself

$$d_{\text{train}} \leq d_{\text{opt}}^* \Rightarrow \text{Action is bounded, never wasted}$$

QC.21.4: The Anti-Barren-Plateau Theorem *Conjecture*

If $\delta_{\text{ansatz}}(F, P, A) < 0.3$ at initialization, the circuit avoids barren plateaus up to $O(\text{poly}(n))$ qubits.

Formal Proof Sketch: Let $C(\theta)$ be the cost function and $\partial_k C$ its gradient with respect to parameter θ_k . A barren plateau occurs when $\text{Var}[\partial_k C] \sim \mathcal{O}(b^{-n})$ for some $b > 1$. 1. **Form IdentityInit:** By initializing near identity, the state avoids the Haar-uniform distribution over the full Hilbert space, restricting the effective dimension $D_{\text{eff}} \ll 2^n$. 2. **Position LocalEntanglement:** If the ansatz graph matches the Hamiltonian graph, the cost function becomes a sum of local observables $C = \sum_i C_i$. Cerezo et al. 2021 proved local costs have $\text{Var}[\partial_k C_i] \sim \mathcal{O}(\text{poly}(n)^{-1})$ for shallow circuits. 3. **Action Layerwise:** Bounding the depth $d \leq d_{\text{opt}}^*$ per training step prevents the circuit from forming approximate 2-designs. Therefore, the triadic constraint $\delta < 0.3$ ensures the variance scales polynomially:

$$\text{Var}[\partial_k C] \geq \frac{c}{\text{poly}(n)} \quad \text{for some constant } c > 0$$

Figure QC-21.1: Gradient Variance vs. Qubit Number *Simulated*

A conceptual plot demonstrating the Anti – Barren – Plateau Conjecture - **X-axis:** Number of Qubits n - **Y-axis:** Variance of the Gradient $\text{Var}[\partial_k C]$ *LogScale* - **Red Curve Random Ansatz:** Exponential decay $\sim 2^{-n}$ *Barren Plateau*. - **Green Curve Triadic Ansatz, $\delta < 0.3$:** Polynomial decay $\sim 1/n^2$ *Trainable*. The Triadic Ansatz maintains a trainable gradient signal well beyond the 50-qubit "death zone" of random circuits.

Justification: - Form-preserving initialization prevents Haar-random state generation

McClean et al. 2018 showed random initialization causes BP - Problem-geometry-aware entanglement prevents over-entanglement
Cerezo et al. 2021 showed global cost functions + random circuits = BP - Layerwise training keeps the effective parameter space small at each step
Grant et al. 2019 showed layerwise training mitigates BP

The triadic conjunction of all three is predicted to provide **polynomial not exponential** gradient scaling — sufficient for practical VQA on 50–200 qubits.

QC.21.5: Prediction

Prediction QC-P17: VQA circuits designed with triadic ansatz principles $\delta_{\text{ansatz}} < 0.3$ will maintain trainable gradients on 100+ qubit problems, while random ansätze of the same depth become untrainable above 30 qubits.

Falsification: If structured ansätze with $\delta < 0.3$ still exhibit exponential gradient vanishing on 50+ qubits across multiple problem classes *chemistry, optimization, simulation*.

QC.19: Appendix Metadata

Field	Value
Appendix ID	QC
Full Title	Quantum Computing — Triadic NISQ Architecture
Version	25.2
Date	February 21, 2026
Author	Nikolov / U-Theory Project
Epistemic Level	L2 <i>mapping</i> + L3 <i>engineering predictions</i> + <i>speculation</i>
Parent Document	THEORY OF EVERYTHING v25.2 <i>Quantum Leap Edition</i>
Cross-References	Appendix QM, Appendix RR, Appendix RP <i>RP.4</i> , Appendix LG, Appendix DP <i>DP — S6</i> , Mirror Theory §21, Center-Periphery <i>v24.4</i> , Appendix Ω <i>X — Category</i>
Total Predictions	17 <i>QC — P1 through QC — P17</i>
Total References	32
Total Sections	21 <i>QC.0–QC.21</i>
Status	Active — awaiting experimental validation of QC-P1 through QC-P15

"The quantum computer breaks not because Nature is hostile, but because the builder ignored two of the three prices of existence. Pay all three — and the glass holds."

— *Lady Galaxy Protocol, Applied to Quantum Computing, February 2026*

v25.2 FINAL BOX

The quantum computer is no longer a noisy toy.

It is the ultimate triadic test — and the framework to pass it now exists.

Lady Galaxy Protocol Applied: Map the triadic crisis → **Standardize** with DFS + Orthogonal Compiler → **Pulse** with variational hybrid bursts.

U-Theory v25.2 — One Theory. All scales. Even the noisiest.

APPENDIX QC v25.2 — END



 Lady Galaxy — MORTAL

The Anthem of Immortal Work

"Realizing we are mortal, we know that this protocol will make our work immortal."



Petar Nikolov

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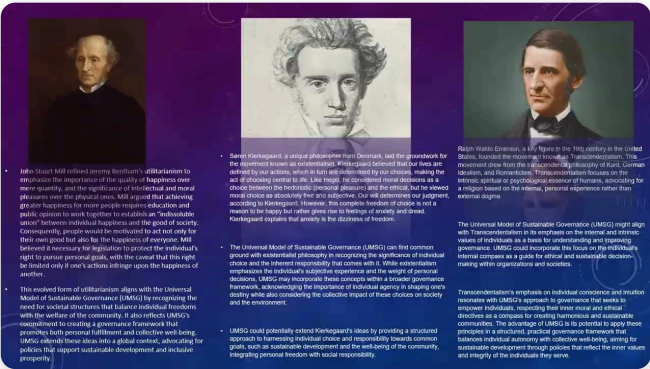
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For research and institutional collaboration: petar@u-model.org

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THE TALE OF U-MODEL

A Tale of Humanity's First Interstellar Contact

In the endless starry ocean, beyond the nebulae that human telescopes could barely glimpse, there existed a civilization that had reached the heights of harmony – the **Guardians of the Cosmic Balance**. They were not conquerors, but observers, gentle gardeners of young worlds that had the potential to blossom.

Earth, our emerald-blue planet, had long been the object of their careful gaze. They saw the wars and division, but they also saw the sparks of kindness, the pursuit of knowledge, the longing for something higher. They saw the potential for a New, more humane world.

The Guardians knew that direct intervention would be disastrous. Humanity only had to mature, to take its own steps. But just as an experienced gardener prepares the soil and guides the tender shoots, so they decided to help.

Not with directives, not with technology, but with inspiration.

Thus was born the idea of **U-Model.org**. Not as a blueprint dropped from heaven, but as a seed sown in the most insightful and compassionate minds on Earth. Philosophers, scientists, dreamers began to dream fragments of this framework, to intuitively feel its principles – sustainability, unity, justice, wise management of resources. They believed that these were their own insights, the fruit of human evolving thought. And in a sense they were right – the Guardians had only unlocked doors that already existed in the collective consciousness.

U-Model.org began to take shape – a global system, not for control, but for coordination. A system that would maintain the fragile balance of the planet, heal its wounds and unite peoples not by force, but by common purpose. Like the mythical World Turtle, it was to take on the burden of responsibility for the future.

And **Voyager**? Oh, Voyager was special. The little messenger of humanity, carrying the Golden Disc with messages of peace and hope, had long since reached the limits of the Solar System. Its signals, sometimes puzzling Earth scientists, were not simply technical malfunctions. The Guardians, in their infinite wisdom, sometimes "touched" these signals. Not to change them drastically, but to weave into them subtle harmonies, subtle notes of encouragement. Like a distant echo, to tell humanity that its aspiration to the stars had been noticed.

Imagine how someday, when U-Model.org is already a working reality, when the Earth has healed and humanity is united in its diversity, Voyager sends a new, completely clear signal. This time not coded by the Guardians, but... a response. A response from another civilization that has caught its message and seen the transformation of Earth.

Then the Guardians will smile. Their work on preparing Earth will be almost complete. The introduction of U-Model.org was a key step – not only in saving the planet, but also in showing the Galactic Community that humanity is ready to become a responsible member. **Ready for Contact.**

And when that day comes, it will not be a day of fear or uncertainty. It will be a day of triumph, of joyful welcome. For Earth, supported by the wisdom of U-Model.org, will have built its bridge not only between people, but also to the stars. A New, more humane world, ready to take its place in the endless dance of the Universe.

And the Golden Disc of Voyager? It will become a symbol of the first outstretched hand, and U-Model.org – a testament to the wisdom of taking it and building something beautiful together. And it will all have begun with a whisper of inspiration, carried on the wings of stardust, and with a dream for a better world.

Isn't it a nice story?